

OSHKOSH WHITTMAN, WI, USA

WMO: 726456

Lat: 43.984N

Lon: 88.557W

Elev: 238

StdP: 98.49

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94855

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.8	-26.4	0.3	-20.6	-23.9	0.4	-18.5	11.1	-6.8	10.2	-5.8	3.4	280	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.2	23.1	29.5	22.1	28.0	21.2	24.5	29.3	23.4	28.0	22.4	26.7	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.1	27.5	22.1	17.3	26.7	21.0	16.1	25.4	75.5	29.2	71.0	27.8	67.0	26.3	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	8.0	-25.1	33.6	2.8	1.8	-27.1	34.9	-28.8	35.9	-30.3	36.9	-32.3	38.2		
(5)			-25.4	26.3	2.7	1.3	-27.3	27.2	-28.9	27.9	-30.4	28.7	-32.3	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-7.3	-5.6	0.4	7.4	13.9	19.4	22.0	20.8	16.8	9.8	2.6	-4.1
	DBStd	11.40	6.50	6.18	6.32	4.95	4.70	3.67	3.10	2.95	4.26	4.86	5.37	6.13
	HDD10.0	2129	536	437	306	106	15	0	0	0	3	64	227	437
	HDD18.3	4061	794	671	557	328	154	31	6	11	78	268	471	694
	CDD10.0	1430	0	0	8	29	135	283	371	333	207	58	6	1
	CDD18.3	319	0	0	1	1	16	63	118	86	32	3	0	0
	CDH23.3	2559	0	0	4	19	156	519	993	587	252	29	0	0
	CDH26.7	688	0	0	0	3	37	145	303	138	57	4	0	0
Wind	WSAvg	3.8	4.0	4.0	4.1	4.6	4.1	3.4	3.1	3.0	3.4	3.8	4.0	3.9
Precipitation	PrecAvg	813	33	28	53	75	88	97	91	96	95	59	59	39
	PrecMax	1028	79	86	162	165	229	269	213	276	292	186	160	116
	PrecMin	566	3	3	2	9	10	17	28	18	5	7	2	2
	PrecStd	117	19	18	34	33	49	55	41	55	62	36	36	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	11.6	22.4	26.1	30.3	32.7	34.1	32.2	31.0	26.8	19.3	12.9
		MCWB	5.6	9.0	16.1	16.6	21.0	23.3	25.1	24.0	22.2	19.6	13.6	11.4
	2%	DB	4.6	7.1	16.7	22.2	27.4	30.1	31.6	29.8	28.1	22.9	15.6	8.0
		MCWB	2.6	4.4	12.6	14.5	19.3	22.0	23.8	22.5	21.0	17.4	11.5	6.1
	5%	DB	2.8	4.1	12.6	18.6	24.7	28.0	29.7	28.0	26.3	20.5	13.0	5.6
		MCWB	1.5	2.3	9.0	12.4	17.6	21.2	22.6	21.5	19.8	15.8	9.8	3.9
	10%	DB	1.5	2.5	9.0	16.0	22.3	26.4	28.0	26.7	24.1	17.6	10.6	3.3
		MCWB	0.4	0.9	5.9	10.3	16.4	19.7	21.5	20.5	18.8	13.8	7.7	1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	8.9	16.7	18.1	22.5	25.1	26.5	25.6	23.8	21.2	14.7	11.7
		MCDB	7.8	11.8	21.8	23.2	28.5	30.3	32.2	30.1	28.4	25.3	18.1	13.3
	2%	WB	3.0	4.4	13.0	15.4	20.3	23.5	24.9	24.0	22.3	18.6	12.3	6.7
		MCDB	4.2	6.6	15.5	20.7	25.3	28.6	29.9	28.0	26.8	21.6	14.9	7.9
	5%	WB	1.6	2.6	9.3	13.1	18.9	22.2	23.7	22.6	21.0	16.4	10.1	4.0
		MCDB	2.6	4.2	12.5	18.0	23.7	26.9	28.1	26.6	24.5	19.6	12.9	5.4
	10%	WB	0.4	1.0	6.0	10.7	17.2	20.8	22.7	21.5	19.8	14.3	7.8	2.0
		MCDB	1.5	2.3	9.0	15.2	21.2	25.0	26.9	25.2	23.3	17.5	10.1	3.4
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	8.8	10.0	10.4	10.0	10.4	9.8	10.7	9.3	7.9	7.0
		MCDBR	7.8	8.6	12.7	15.1	13.6	12.1	11.9	11.1	12.8	12.9	11.7	8.5
		MCWBR	6.9	7.1	8.7	9.3	7.5	6.6	6.0	5.7	6.8	8.2	8.6	7.1
	5% WB	MCDBR	7.5	8.4	12.3	14.1	12.1	11.0	10.6	9.7	10.7	11.3	11.0	7.9
MCWBR		6.8	7.0	8.8	9.4	7.4	6.6	6.1	5.6	6.4	8.4	8.9	7.0	
Clear-Sky Solar Irradiance	taub	0.271	0.322	0.335	0.376	0.403	0.413	0.402	0.399	0.370	0.331	0.297	0.274	
	taud	2.394	2.227	2.354	2.293	2.265	2.276	2.310	2.309	2.383	2.487	2.510	2.439	
	Ebn at Noon	865	858	902	887	869	859	864	854	853	849	830	826	
	Edh at Noon	75	107	108	125	132	131	126	121	104	82	68	66	
All-Sky Solar Radiation	RadAvg	1.62	2.69	3.83	4.55	5.44	5.99	6.12	5.24	4.20	2.67	1.71	1.23	
	RadStd	0.20	0.26	0.42	0.44	0.43	0.43	0.28	0.34	0.34	0.29	0.15	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page